Urinary Introduction

The main role of the urinary system is to control the body's fluids yet has a minimal role for its excretory function.
An adult has around **35-50** litres of fluid, around 15 litres of this is extracellular fluid. The remaining intracellular fluid composition and volume is controlled by the extracellular fluid. The urinary system acts to control the extracellular fluid, so can affect not just blood pressure, but also tissue fluid and the cells functions.

Of the 15 litres of extracellular fluid, 3 litres is plasma and it is these 3 litres of fluid that the kidneys act upon. Affecting this causes changes to intracellular compositions and volumes. Due to the semipermeable nature of the cell membranes, water can move freely across them and is affected by the solute concentration (the osmolarity). Normal osmolarity is around 300 milliosmoles. Failure to control osmolarity will cause cellular changes and eventual cell function failure.
The kidneys also have a part in regulating acid-base balance which is kept between 7.35-7.45. The kidneys control the hydrogen carbonate concentrations, which is described in more detail later.
Kidneys also remove toxic metabolites from the blood, such as urea and creatinine.
So overall the kidney will:

- Control cell volume
- Control pH partially (with respiratory system)
- Control osmolarity
- Excrete waste products

The kidneys filter a large amount of fluid, around 180 litres per day, and recover nearly all of it.

However, this does mean there is a large metabolic activity of the kidney and it requires around 25% of the cardiac output to function correctly. This means the kidney are a vulnerable organ and have a very high risk of damage, even with slight ischemia.

Each kidney ways around 150g, within each kidney, there is a structure that allows the urine that is produced to pass down into the ureters, then subsequently into the bladder.

Each kidney is surrounded by a fibrous capsule which is further surrounded by perinephric fat. The renal cortex is the outer zone of the kidney and the renal medulla is the inner zone, which is made up of renal pyramids. The cortex contains all the glomeruli, and the proximal and distal convoluted tubule, and the medulla contains loop of Henle, vasa recta, and the final portions of the collecting ducts.

The collecting duct of every nephron drains into the renal papillas, which are found at the apex of the renal pyramids. The renal papilla drain into the minor calyx which then drain into the major calyx; which in turn drain into the renal pelvis.
There are usually 2-3 major calyces and 2-3 minor calyces per major calyces. The renal pelvis eventually becomes the ureters.

The blood supply to the kidney comes from the renal artery and renal vein, which enter the kidney at the renal hilum. The renal arteries divide into interlobar arteries which penetrate into the renal cortex, which then divide into numerous arcuate arteries. These then divide into arcuate arteries which run along the boundary of the cortex and medulla and then divide into interlobular arteries.
The interlobular arteries then feed into the afferent arterioles.

The main hormones that act on the kidney are:

- Antidiuretic hormone (ADH) – Peptide which is released from the **posterior pituitary gland** and promotes **water reabsorption** in the **collecting duct**
- Aldosterone – Steroid hormone that is produced by the **adrenal cortex** and promotes **sodium and water reabsorption** in the **collecting ducts**
- Natriuretic peptides (NPs) – These are produced by **cardiac myocytes** and promote **sodium and water excretion** in the **collecting ducts**
- Parathyroid hormone (PTH) – This protein is produced by the **parathyroid gland** and promotes **phosphate excretion and calcium reabsorption, as well as vitamin D production**.

The main hormones produced by the kidney are:

- **Renin – Protein released by juxtaglomerular apparatus** and has part in RAAS system (**Renin-Angiotensin-Aldosterone System).** Stimulation of production of angiotensin II causes sodium and water reabsorption and also as a potent vasoconstrictors (rises BP)
- **Vitamin D – Steroid hormone** that is metabolised in the kidney to produce (**1,25-dihydroxyvitamin D**) which promotes calcium and phosphate absorption at the gut.
- **Erythropoietin – Protein produced** by the kidney which promotes erythropoeisis in bone marrow in response to hypoxic conditions.
- **Prostaglandins** – Produced by the kidney and have numerous effects.